Fiscal Year:	FY 2010	**Task Last Updated:**	FY 05/03/2010
PI Name:	Barcellos-Hoff, Mary Helen Ph.D.		
Project Title:	NSCOR: The contribution of non-targeted effects in HZE cancer risk		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mary.barcellos-hoff@ucsf.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	415-476-8091
Organization Name:	University of California, San Francisco		
PI Address 1:	Department of Radiation Oncology		
PI Address 2:	2340 Sutter Street		
PI Web Page:			
City:	San Francisco	**State:**	CA
Zip Code:	94115	**Congressional District:**	12
Comments:	NOTE: Former affiliation was Lawrence Berkeley National Laboratory; moved to NYU in August 2008. Moved to UCSF September, 2015		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 NSCOR Space Radiation NNJ08ZSA003N
Start Date:	07/01/2009	**End Date:**	06/30/2014
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Costes, Sylvain (Lawrence Berkeley National Laboratory) Demaria, Sandra (NYU Medical Center) Mao, Jian-Hua (Lawrence Berkeley National Laboratory) Balmain, Allan (University of California San Francisco)		
Grant/Contract No.:	NNX09AM52G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Clear mechanistic understanding of the biological processes elicited by high LET radiation increase cancer risk will reduce the uncertainties in assessing the risk of space flight. Classic targeted radiation effects like cell kill and mutation are proportional to dose and their LET dependence is described by relative biological effectiveness. However recent data suggests that high LET radiation affects multicellular processes that promote cancer. We propose that radiation carcinogenesis results from two classes of ionizing radiation effect: targeted effects that alter genomic sequence and non-targeted effects (NTE) that alter phenotype and multicellular interactions. Since NTE are not proportional to dose, new approaches are required to assess the impact of NTE, which alter gene expression, cell signaling, genomic stability and cell phenotypes, on carcinogenesis. It is unknown to what extent NTE contribute to the significantly greater carcinogenic effect of high LET radiation. We hypothesize that NTE promotes epithelial cancer. Project 1 will test use novel transgenic and humanized mouse mammary models to evaluate the relative contribution of radiation targeted and NTE to carcinogenesis as a function of radiation quality and of age a radiation chimera model. We will monitor inflammation as a key process affected by radiation NTE and conduct a systems biology analysis of mammary gland. Project 2 will employ new systems genetic analysis of cancer susceptibility in radiation sensitive organs to define critical regulatory networks that impact the relative carcinogenic effectiveness of high LET radiation. Project 3 will use Monte Carlo simulations of targeted radiation effects and agent based modeling of cell interactions to model the relative contributions of targeted and NTE to carcinogenesis. This NSCOR will provide important data on radiation quality dependence of targeted and non-targeted radiation effects and the genetic determinants of cancer susceptibility, which together will used to reduce uncertainties in NASA radiation risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	First, radiation can be considered an experimental probe that perturbs tissue function. Studying these perturbations is a means to define critical biological response to stress. Second, we will develop a comprehensive data base of normal tissue and irradiated tissue by analyzing different organs in each mouse. This data base will become a public resource. Third, our studies will identify genetic predisposition to radiation associated cancer, and may thus help protect humans who have the greatest risk of cancer.
Task Progress:	The first year of the funding was devoted in part to staffing and training the three projects encompassed in this NASA Specialized Center of Research. The PI and co-investigators met three times to discuss the research strategy based on the revised scope of the funding. Three postdoctoral fellows and two staff members were recruited to execute the proposed projects, each of whom completed training for BNL and NSRL access and use. In the first year, we have completed two campaigns at NSRL using two particle beams consisting of approximately 300 mice. Six organs were collected at three or more time points for analysis. Mammary gland tumors will be monitored for the next year in mice receiving p53 null epithelial transplants after irradiation and compared to controls irradiated with low LETat BNL.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Articles in Peer-reviewed Journals	Costes SV, Chiolo I, Pluth JM, Barcellos-Hoff MH, Jakob B. "Spatiotemporal characterization of ionizing radiation induced DNA damage foci and their relation to chromatin organization." Mutation Research 2010 Apr-Jun;704(1-3):78-87. Epub 2010 Jan 8. Review. PubMed PMID: 20060491 , Apr-2010